Chenyang LIU

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Education

Duke University — Durham, USA
PhD in Computer Science, advised by Prof. Matthew Lentz and Prof. Kartik Nayak — *Aug. 2024 – Present*

Shanghai Jiao Tong University — Shanghai, China
Bachelor of Engineering in Information Security, GPA: 90.33/100, Rank: 3/86 — *Sep. 2020 – June 2024*

Research Interests

My research lies at the intersection of distributed systems, blockchain, and security. I design and implement Byzantine Fault-Tolerant (BFT) consensus protocols that are both secure and efficient; my current work explores signature-free BFT protocols that remain secure in the post-quantum era. I also build confidential computing systems on Trusted Execution Environments (TEEs), from decentralizing Ethereum's proposer-builder separation (PBS) to mitigating side-channel attacks.

Publications & Manuscripts

Quintus: Two-round Good-case Information Theoretic BFT for $n = 5f + 1$ — In Submission
Chenyang Liu, *Dahlia Malkhi, Kartik Nayak, Nibesh Shrestha*

DPaaS: Improving Decentralization and Trustworthiness in Ethereum PBS — In Submission
Chenyang Liu, *Arnav Jindal, Ittai Abraham, Matthew Lentz, Kartik Nayak*

A Formal Approach to Multi-Layered Privileges for Enclaves — NDSS 2025
Ganxiang Yang, ***Chenyang Liu***, *Zhen Huang, Guoxing Chen, Hongfei Fu, Yuanyuan Zhang, Haojin Zhu*

Research Experience

Graduate Research Assistant — Aug. 2024 – *Present*
Duke University — *Durham, USA*

DPaaS: Decentralized Proposer-as-a-Service for Ethereum PBS

- **Protocol Design**: Formalized the security properties of Ethereum PBS and designed a side-car protocol that outsources the combined roles of relay and proposer to a group of TEEs.
- **New Primitive Crab-BB**: Designed a generalized Byzantine Broadcast protocol with censorship resistance and backed availability for arbitrary applications, including Ethereum PBS.
- **Implementation and Evaluation**: Built a Rust prototype deployed on AWS, Azure, and GCP confidential VMs (AMD SEV-SNP, Intel TDX), showing 3% higher MEV earnings than relays.

Signature-Free BFT Consensus for the Post-Quantum Era

- **Quintus**: PBFT-style information-theoretic BFT protocols for $n = 5f + 1$ with two-round good-case commit (matching signature-based protocols), $O(n^2)$ messages, and optimistic responsiveness.

Undergraduate Research Assistant — Feb. 2023 – July 2024
Shanghai Jiao Tong University, advised by Prof. Guoxing Chen — *Shanghai, China*

Closing Cache Side Channels via Minimal Cache Fingerprints

- **Cache-Set Flattening**: Designed the first flattening-based defense against cache side channels on Intel SGX, confining all sensitive code and data to a single cache set so that access traces reveal no secrets.
- **Compiler and Driver Support**: Built LLVM backend passes that split enclave code/data into 64-byte blocks and instrument memory accesses, plus an SGX driver for cache-color-aware page allocation.
- **Verifiable OS Contract**: Proposed *contrived self-eviction*, a timing-based test that lets the enclave verify the untrusted OS's page allocation on both inclusive and non-inclusive caches.
- **Evaluation**: Defeated Prime+Probe attacks on OpenSSL AES and GnuPG ElGamal with 3.25× overhead on SGX-nbench; optimized contract verification to cut wolfSSL overhead from 12× to 1.24×.

Formally Verified Privileged Enclaves

- **Proof-driven Design**: Architected and formally verified a novel service framework for Trusted Execution Environments (TEE) to ensure robust security guarantees.
- **Case Studies and Evaluation**: Implemented three case studies and evaluated the performance of the proposed design on the Penglai RISC-V TEE platform.

Teaching & Service

Teaching Assistant: Intro. to Operating Systems (Spring 2025); Operating Systems (Fall 2025)

Artifact Evaluation Committee: ACM CCS 2027

Selected Honors & Awards

Flashbots MEV Fellowship Grant: As student researcher 2025

Outstanding Graduate: Shanghai Jiao Tong University 2024

Zhiyuan Honor Awards: Top 5% of students in Zhiyuan Honor Program 2021 – 2023

First Class Network Security Awards: For contributions to security research 2023

Academic Progress Awards: For outstanding academic progress (3/88) 2022

Skills

Programming: Rust, C/C++, Python, x86/RISC-V Assembly, SQL, JavaScript, MATLAB

TEEs: Intel SGX, AMD SEV-SNP, Intel TDX, Penglai (RISC-V)

Frameworks & Tools: Git, LaTeX, Docker, LLVM, QEMU, Django, Google/AWS/Azure Cloud

Languages: Chinese (Native), English